

diff(1) — Linux manual page

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 DIFF(1)

User Commands

DIFF(1)

NAME [top](#)

diff – compare files line by line

SYNOPSIS [top](#)

diff [*OPTION*]... *FILES*

DESCRIPTION [top](#)

Compare *FILES* line by line.

Mandatory arguments to long options are mandatory for short options too.

- normal**
output a normal diff (the default)
- q, --brief**
report only whether files differ
- s, --report-identical-files**
also report if two files are the same
- c, -C NUM, --context[=*NUM*]**
output NUM (default 3) lines of copied context
- u, -U NUM, --unified[=*NUM*]**
output NUM (default 3) lines of unified context
- e, --ed**
output an ed script
- n, --rcs**
output an RCS format diff
- y, --side-by-side**
output in two columns

-W, --width=NUM
output at most NUM (default 130) print columns

--left-column
output only the left column of common lines

--suppress-common-lines
do not output common lines

-p, --show-c-function
show which C function each change is in

-F, --show-function-line=RE
show the most recent line matching RE

--label LABEL
use LABEL instead of file name and timestamp (can be repeated)

-t, --expand-tabs
expand tabs to spaces in output

-T, --initial-tab
make tabs line up by prepending a tab

--tabsize=NUM
tab stops every NUM (default 8) print columns

--suppress-blank-empty
suppress space or tab before empty output lines

-l, --paginate
pass output through 'pr' to paginate it

-r, --recursive
recursively compare any subdirectories found

--no-dereference
don't follow symbolic links

-N, --new-file
treat absent files as empty

--unidirectional-new-file
treat absent first files as empty

--ignore-file-name-case
ignore case when comparing file names

--no-ignore-file-name-case
consider case when comparing file names

-x, --exclude=PAT
exclude files that match PAT

-X, --exclude-from=FILE
exclude files that match any pattern in FILE

-S, --starting-file=FILE

start with FILE when comparing directories

- `--from-file=FILE1`
compare FILE1 to all operands; FILE1 can be a directory
- `--to-file=FILE2`
compare all operands to FILE2; FILE2 can be a directory
- `-i, --ignore-case`
ignore case differences in file contents
- `-E, --ignore-tab-expansion`
ignore changes due to tab expansion
- `-Z, --ignore-trailing-space`
ignore white space at line end
- `-b, --ignore-space-change`
ignore changes in the amount of white space
- `-w, --ignore-all-space`
ignore all white space
- `-B, --ignore-blank-lines`
ignore changes where lines are all blank
- `-I, --ignore-matching-lines=RE`
ignore changes where all lines match RE
- `-a, --text`
treat all files as text
- `--strip-trailing-cr`
strip trailing carriage return on input
- `-D, --ifdef=NAME`
output merged file with '#ifdef NAME' diffs
- `--GTYPE-group-format=GFMT`
format GTYPE input groups with GFMT
- `--line-format=LFMT`
format all input lines with LFMT
- `--LTYPE-line-format=LFMT`
format LTYPE input lines with LFMT

These format options provide fine-grained control over the output

of diff, generalizing `-D/--ifdef`.

LTYPE is 'old', 'new', or 'unchanged'.
GTYPE is LTYPE or 'changed'.

GFMT (only) may contain:

%< lines from FILE1

`%>` lines from FILE2

`%=` lines common to FILE1 and FILE2

`%[-][WIDTH][.[PREC]]{doxX}LETTER`
printf-style spec for LETTER

LETTERs are as follows for new group, lower case for old group:

F first line number

L last line number

N number of lines = L-F+1

E F-1

M L+1

`%(A=B?T:E)`
if A equals B then T else E

LFMT (only) may contain:

`%L` contents of line

`%l` contents of line, excluding any trailing newline

`%[-][WIDTH][.[PREC]]{doxX}n`
printf-style spec for input line number

Both GFMT and LFMT may contain:

`%%` %

`%c'C'` the single character C

`%c'W000'`
the character with octal code 000

C the character C (other characters represent themselves)

-d, --minimal
try hard to find a smaller set of changes

--horizon-lines=NUM
keep NUM lines of the common prefix and suffix

--speed-large-files
assume large files and many scattered small changes

--color[=WHEN]
color output; WHEN is 'never', 'always', or 'auto'; plain
--color means **--color='auto'**

--palette=PALETTE
the colors to use when **--color** is active; PALETTE is a
colon-separated list of terminfo capabilities

--help display this help and exit

-v, --version
output version information and exit

FILES are 'FILE1 FILE2' or 'DIR1 DIR2' or 'DIR FILE' or 'FILE DIR'. If **--from-file** or **--to-file** is given, there are no restrictions on FILE(s). If a FILE is '-', read standard input. Exit status is 0 if inputs are the same, 1 if different, 2 if trouble.

AUTHOR [top](#)

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REPORTING BUGS [top](#)

Report bugs to: bug-diffutils@gnu.org
GNU diffutils home page: <<https://www.gnu.org/software/diffutils/>>
General help using GNU software: <<https://www.gnu.org/gethelp/>>

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SEE ALSO [top](#)

[wdiff\(1\)](#), [cmp\(1\)](#), [diff3\(1\)](#), [sdiff\(1\)](#), [patch\(1\)](#)

The full documentation for **diff** is maintained as a Texinfo manual. If the **info** and **diff** programs are properly installed at your site, the command

info diff

should give you access to the complete manual.

COLOPHON [top](#)

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Pages that refer to this page: [cmp\(1\)](#), [diff3\(1\)](#), [diffman-git\(1\)](#), [gendiff\(1\)](#), [git-config\(1\)](#), [git-diff\(1\)](#), [git-diff-files\(1\)](#), [git-diff-index\(1\)](#), [git-diff-pairs\(1\)](#), [git-diff-tree\(1\)](#), [grep\(1\)](#), [patch\(1\)](#), [quilt\(1\)](#), [sdiff\(1\)](#), [suffixes\(7\)](#)

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